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Generic Virtual Synchronous Machine Grid-Forming Inverter Model—REGFM_B1

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Acknowledgement

- I would like to thank all the dedicated contributors for developing, improving, and implementing the REGFM_B1 model

Contributors:

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Sebastian Achilles	General Electric
Deepak Ramasubramanian	Electric Power Research Institute
Philip Hart	General Electric
Shruti Rao	General Electric
Wenzong Wang	Electric Power Research Institute
Quan Nguyen	Pacific Northwest National Laboratory
Jinho Kim	Pacific Northwest National Laboratory
Qian Zhang	Electric Power Research Institute
Hanchao Liu	General Electric
Pedro Arsuaga Santos	General Electric
James Weber	PowerWorld
Juan Sanchez	General Electric
Mengxi Chen	General Electric
Jayapalan Senthil	Siemens PTI
Pouyan Pourbeik	Power and Energy, Analysis, Consulting and Education (PEACE) PLLC
Udoka Nwaneto	Pacific Northwest National Laboratory
Jeff Bloemink	PowerTech Labs
Song Wang	Portland General Electric
Doug Tucker	Western Electricity Coordinating Council
Songzhe Zhu	GridBright

Timeline of VSM GFM Model REGFM_B1

- The original version of virtual synchronous machine (VSM) grid-forming inverter (GFM) model specification was proposed to WECC MVS in **February 2023** by PNNL, GE, and EPRI. SGRE also provided inputs to the model specification
- The specification was renamed to REGFM_B1 and has been revised multiple times based on the discussions with all the contributors in the past months
- The REGFM_B1 specification was approved by WECC MVS on **Jan. 25th, 2024**
- The REGFM_B1 model has been implemented in PSS/E, PSLF, PowerWorld, and TSAT, and the model benchmarking is complete in **May 2024**



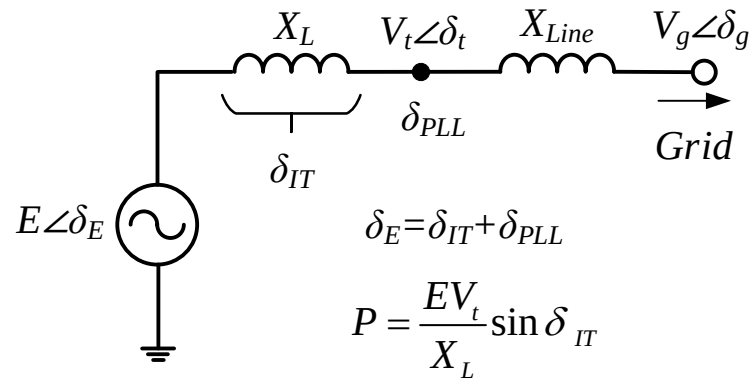
Virtual Synchronous Machine Grid-Forming Inverter Model Specification (REGFM_B1)

Contributors:

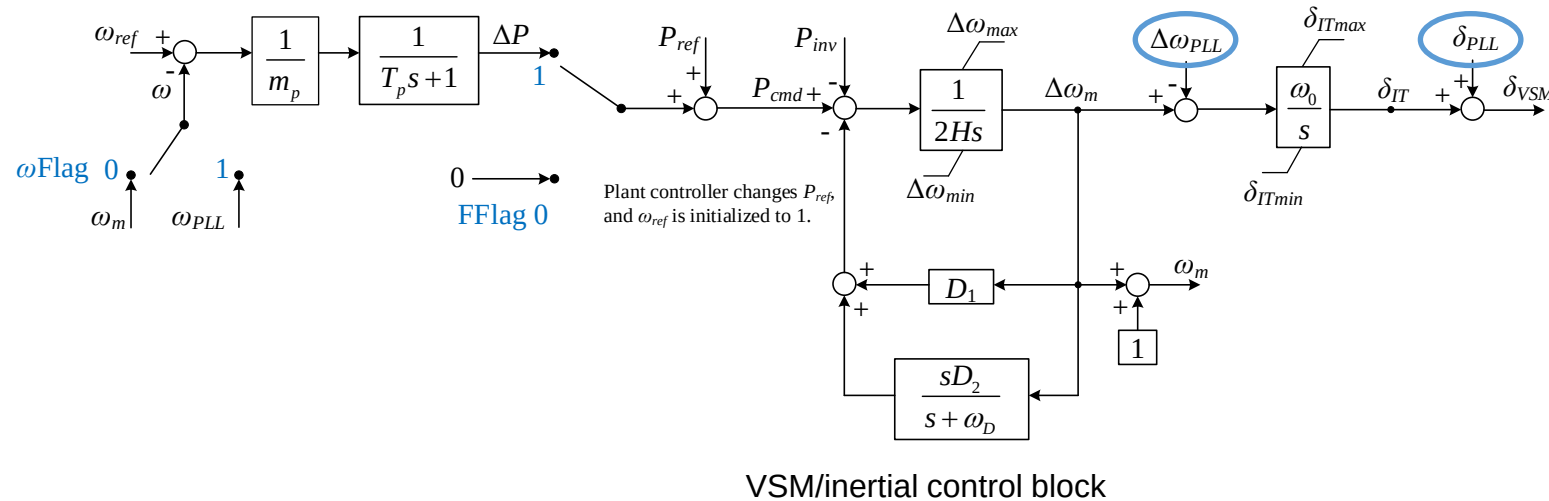
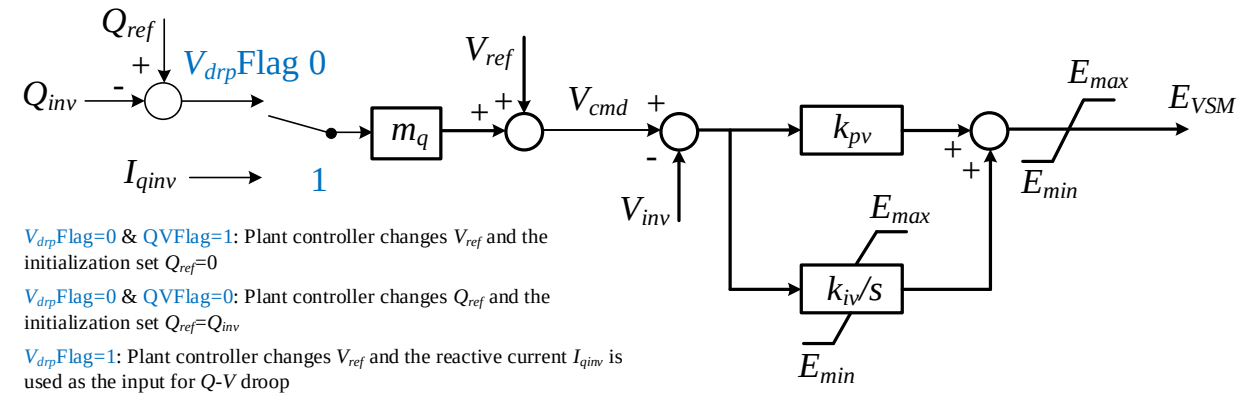
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REGFM_B1 Specification

- The REGFM_B1 specification includes the voltage control block and the VSM/inertial control block
- A PLL is used to obtain the angle of the terminal voltage, and the VSM/inertial block controls the angle difference δ_{IT}
- The use of PLL in the VSM GFM is different from its use in GFLs
- The PLL can be considered as not existent during normal operation (when $\delta_{ITmin} < \delta_{IT} < \delta_{ITmax}$)

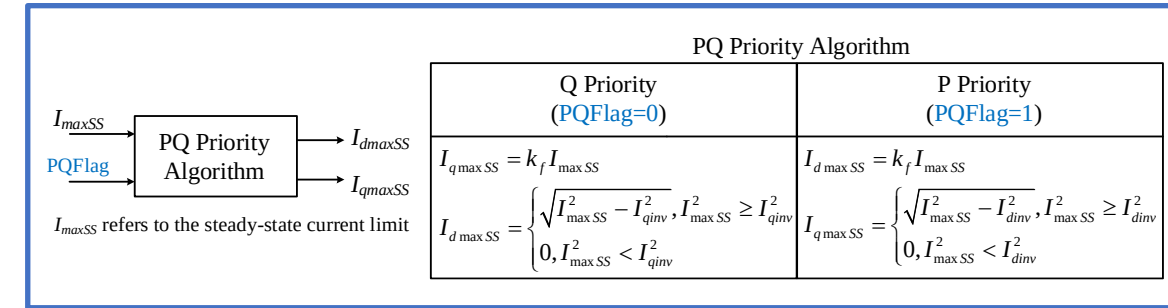


Voltage source behind impedance

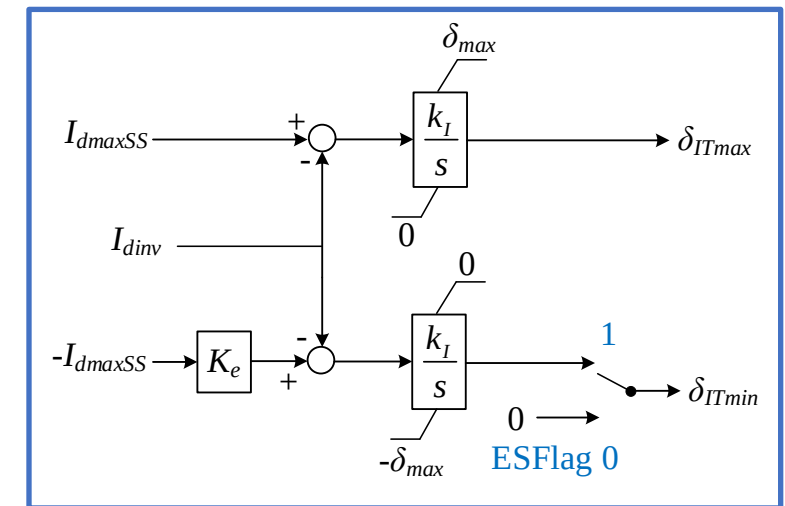
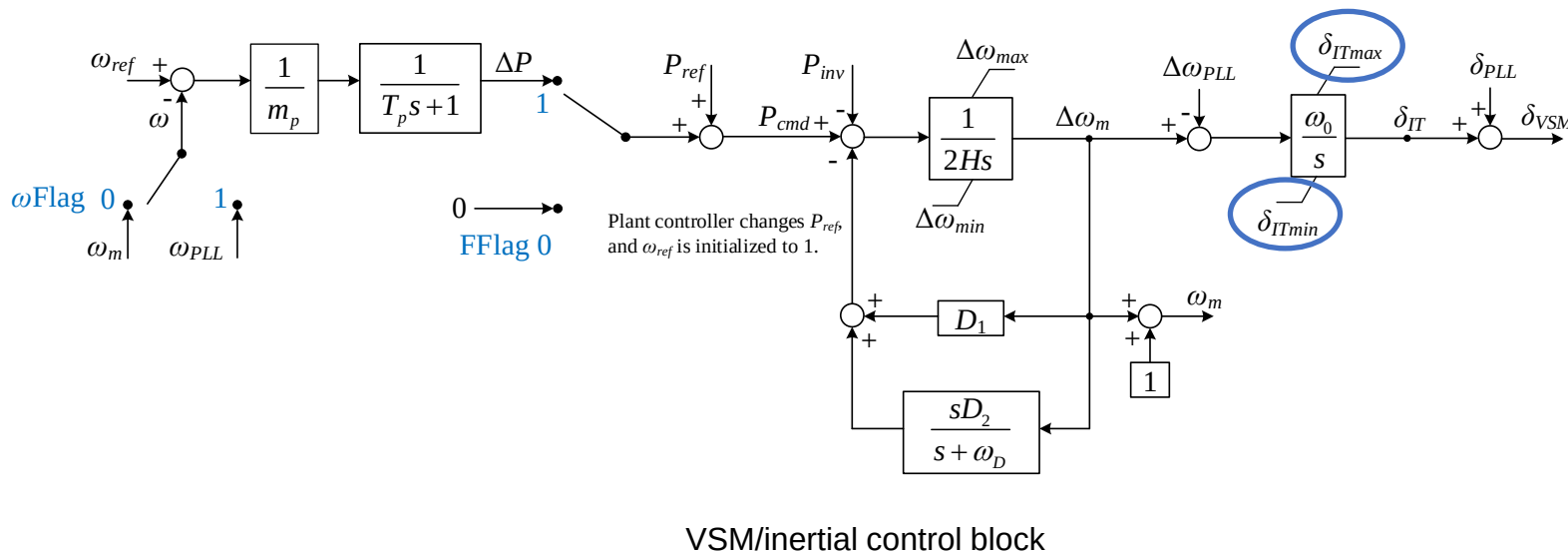


Steady State Current Limiting in REGFM_B1

- The REGFM_B1 model specification includes the steady state **active and reactive current limiting** controls
- The steady state current limiting control utilizes the PQ priority algorithm
- The active current I_{dinv} can be limited by regulating the saturation limits of δ_{IT} of the VSM/inertial control block

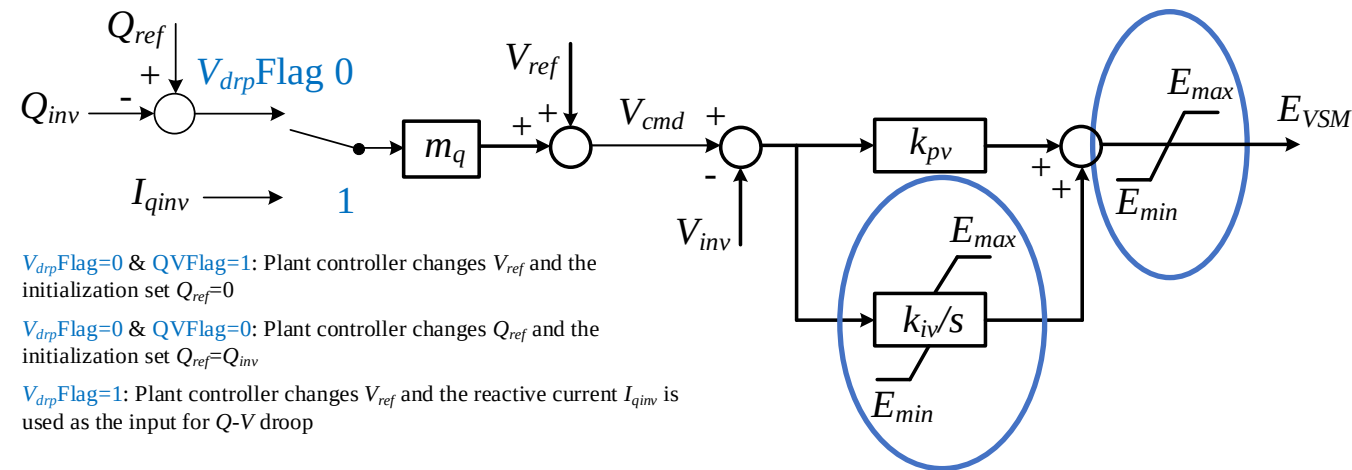


PQ Priority Algorithm to determine I_{dmaxSS} and I_{qmaxSS}



Steady State Current Limiting in REGFM_B1

- The reactive current I_{qinv} can be limited by regulating the saturation limits E_{max} and E_{min} of the voltage control block



Voltage control block

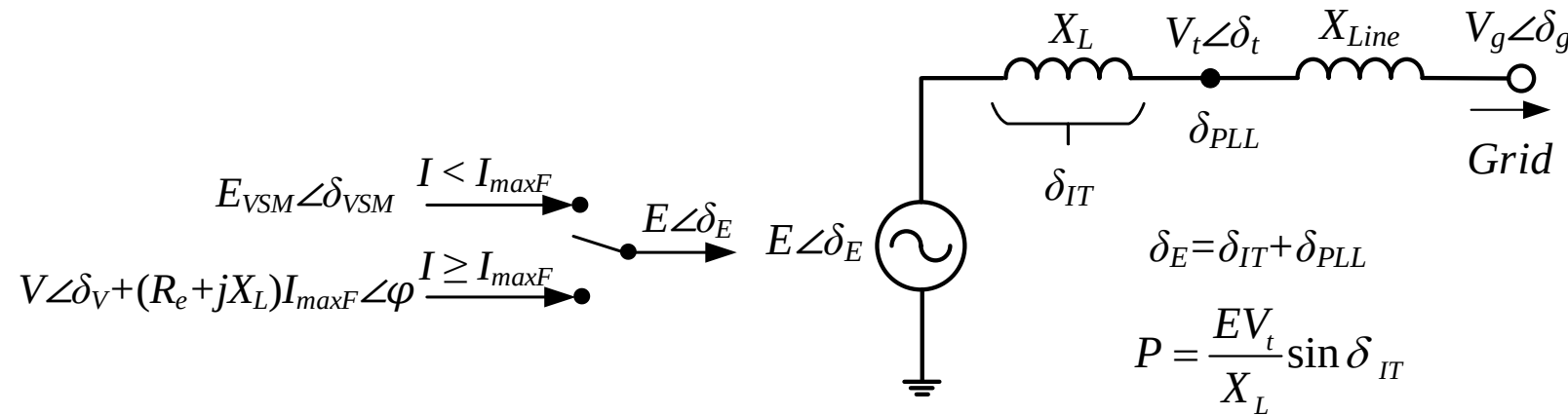
$$E_{min} = \sqrt{(V_{inv} - I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

$$E_{max} = \sqrt{(V_{inv} + I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

Reactive current limiting

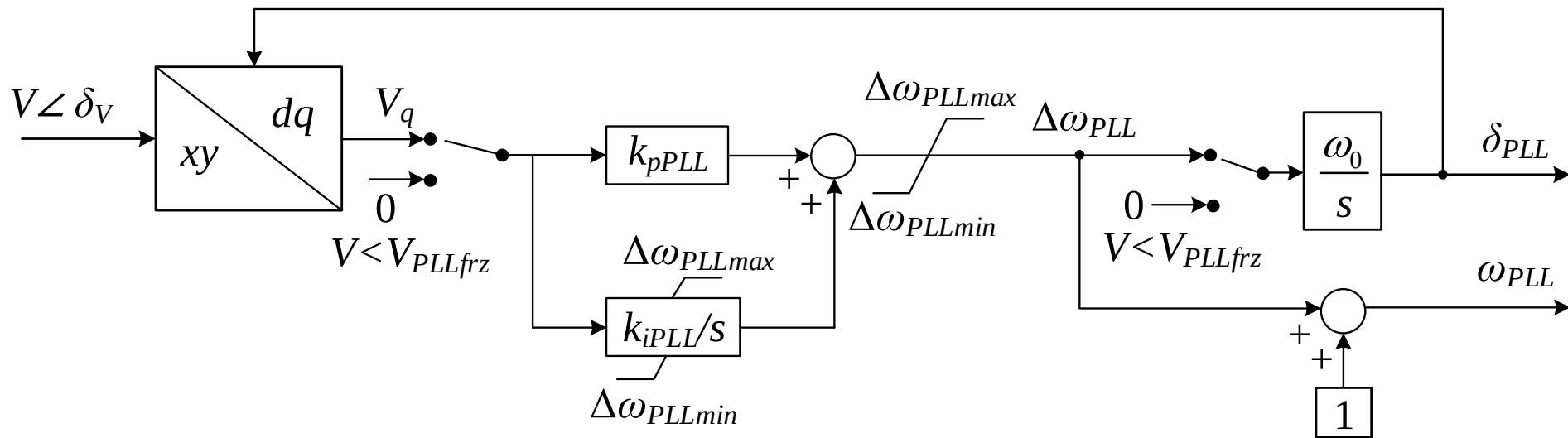
Transient Current Limiting of VSM GFM

- A transient current limiting function is used to limit the output current algebraically at the beginning of a fault, which is the same as used by REGFM_A1



Updates on the PLL Block

- A freeze function was added to the PLL block diagram. When the actual terminal voltage magnitude V (not the filtered value) is less than V_{PLLfrz} , the PLL will do the following:
 - Input to the PI block ($K_{pPLL} + K_{iPLL}/s$) is set to 0
 - The δ_{PLL} state (output of ω_0/s) is frozen



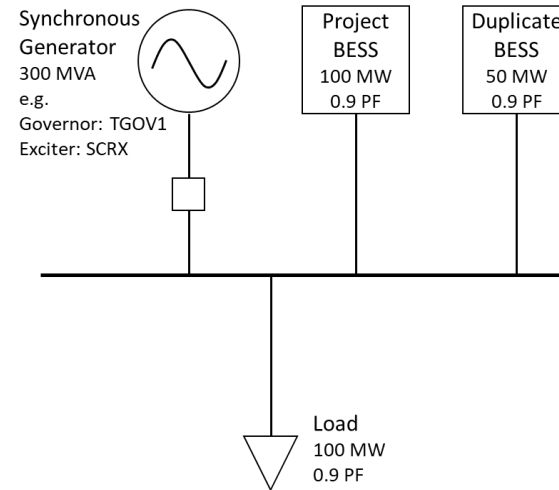
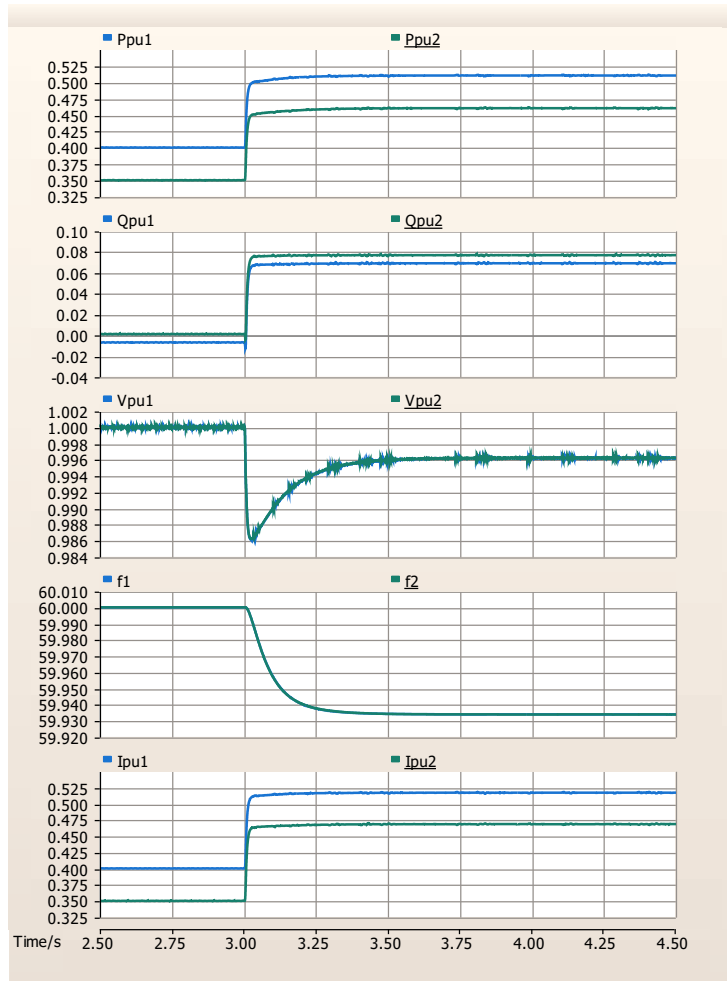
PLL block diagram

PSCAD and PSS/E Simulation Results of REGFM_B1

- Both PSS/E and PSCAD models were developed based on the REGFM_B1 model specification
- The PSCAD model is an IGBT-based detailed model that includes additional PWM control, and the transient current limiting is implemented at the PWM control layer
- The transient current limiting in PSS/E is implemented algebraically
- All parameters were set the same in the PSS/E and PSCAD models
- An islanding test was firstly done for a two-GFM system, and then more testing was done on a single-GFM infinite-bus system
- 18 variables were compared for all cases

Islanding

- After disconnecting from the grid, the two GFM's can maintain the stable operation of the grid
- The REGFM_B1 model can pass the three test cases defined in the NERC GFM functional specification



Source: NERC GFM report

NERC
NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

White Paper: Grid Forming Functional Specifications for BPS-Connected Battery Energy Storage Systems

September 2023

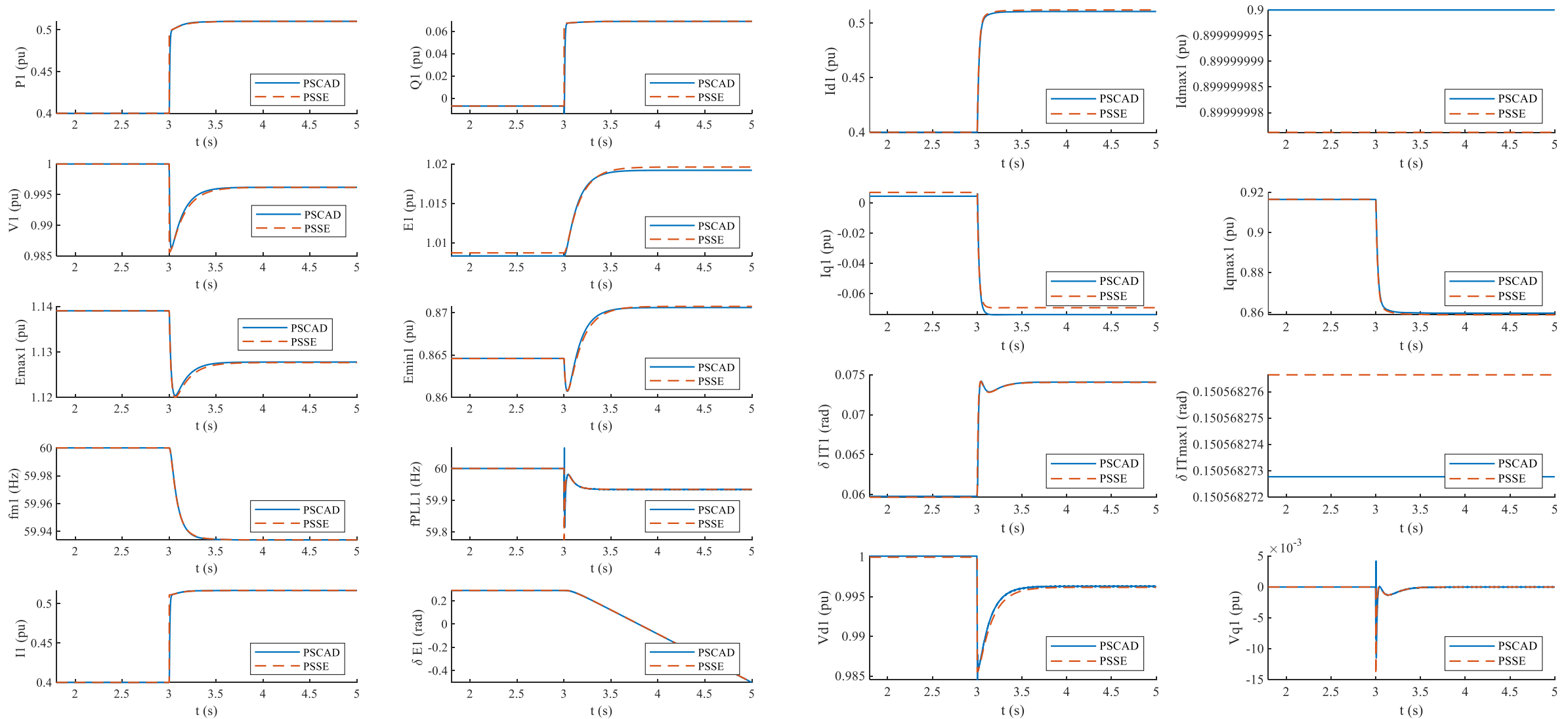
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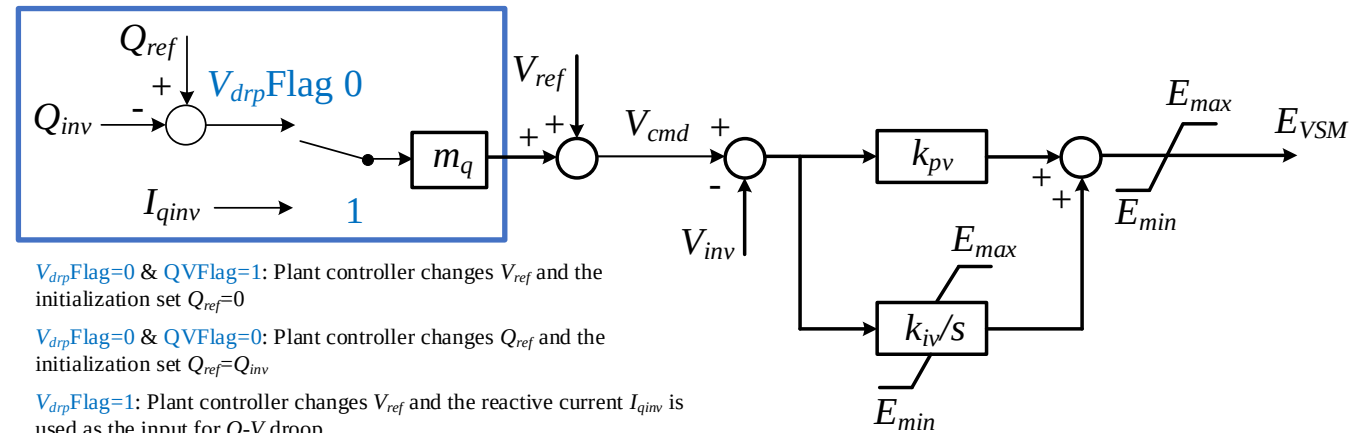
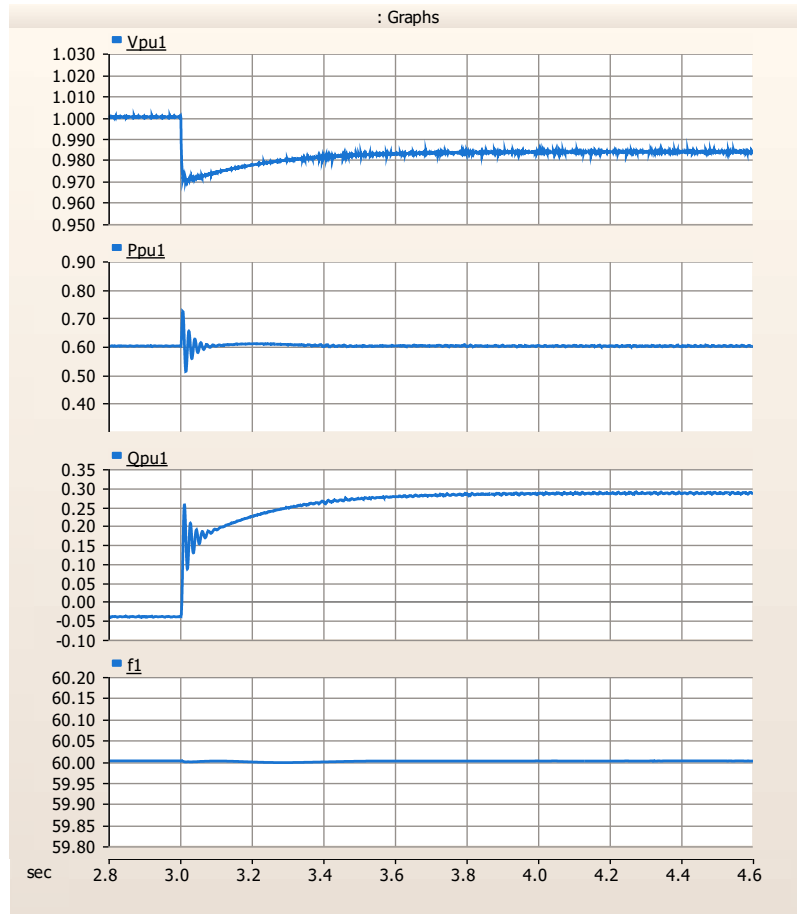
Islanding (PSS/E and PSCAD comparison)

- 18 variables were compared and achieved a good level of alignment



5% Voltage Drop

- When the grid voltage drops, the VSM GFM increases the reactive power Q
- The steady state Q follows the Q-V droop



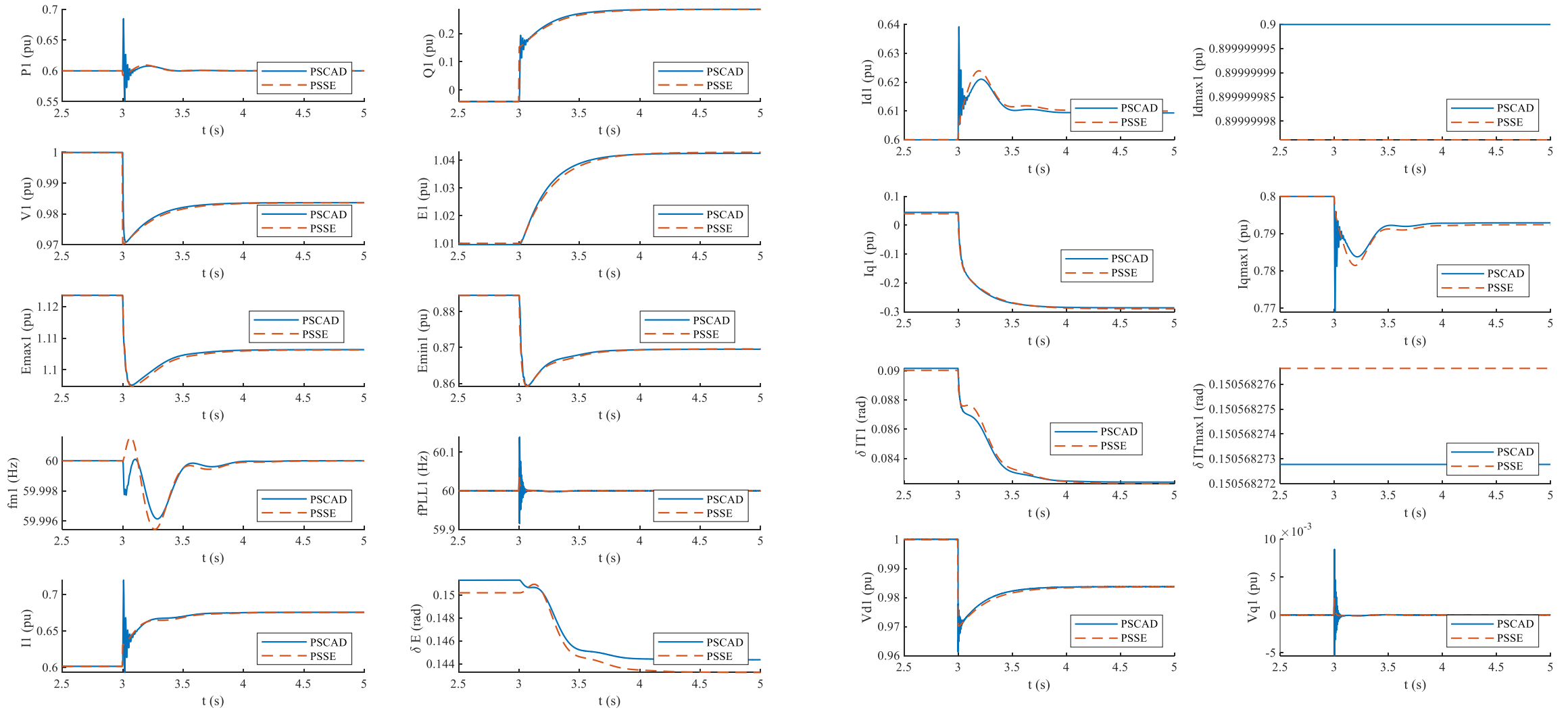
$V_{drp}Flag=0$ & $QVFlag=1$: Plant controller changes V_{ref} and the initialization set $Q_{ref}=0$

$V_{drp}Flag=0$ & $QVFlag=0$: Plant controller changes Q_{ref} and the initialization set $Q_{ref}=Q_{inv}$

$V_{drp}Flag=1$: Plant controller changes V_{ref} and the reactive current I_{qinv} is used as the input for Q-V droop

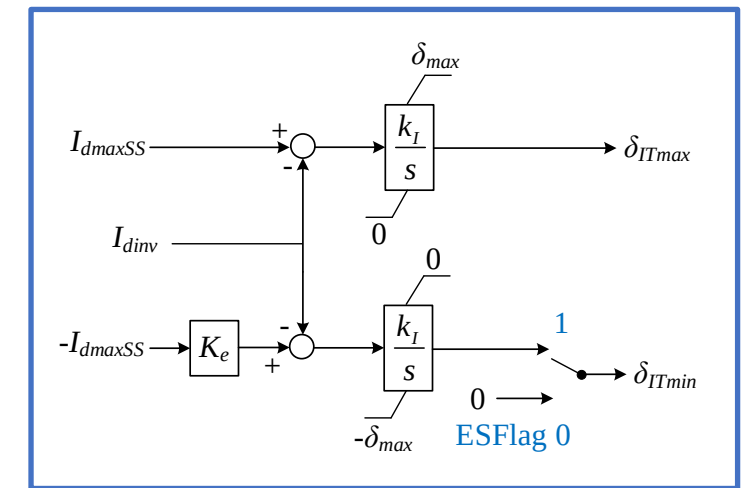
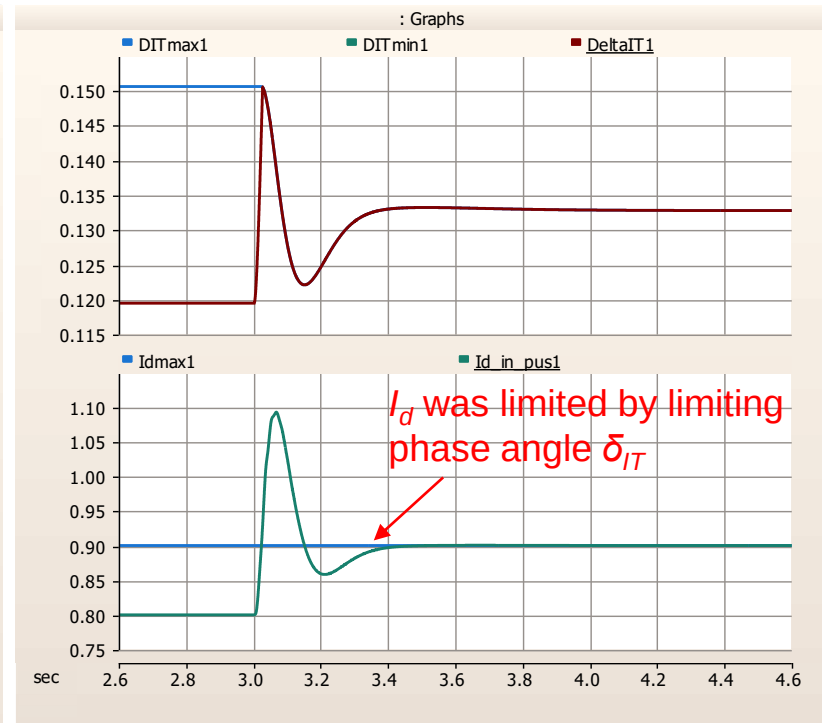
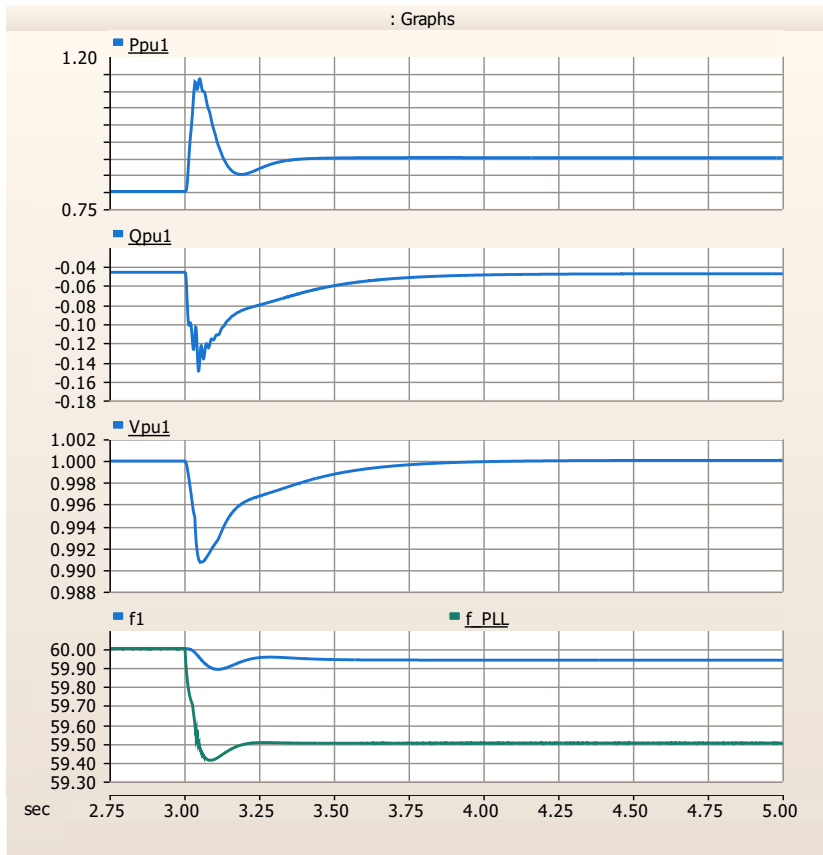
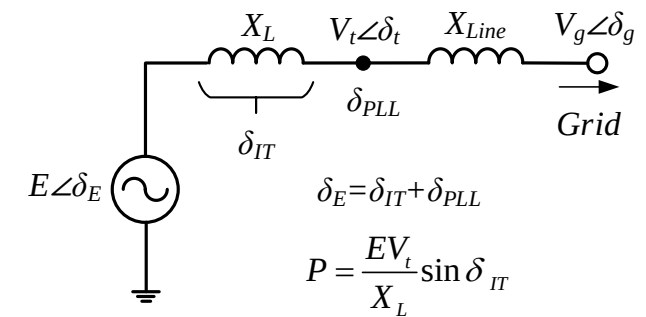
5% Voltage Drop (PSS/E and PSCAD comparison)

- 18 variables were compared and achieved a good level of alignment



Frequency Drop (59.5 Hz)

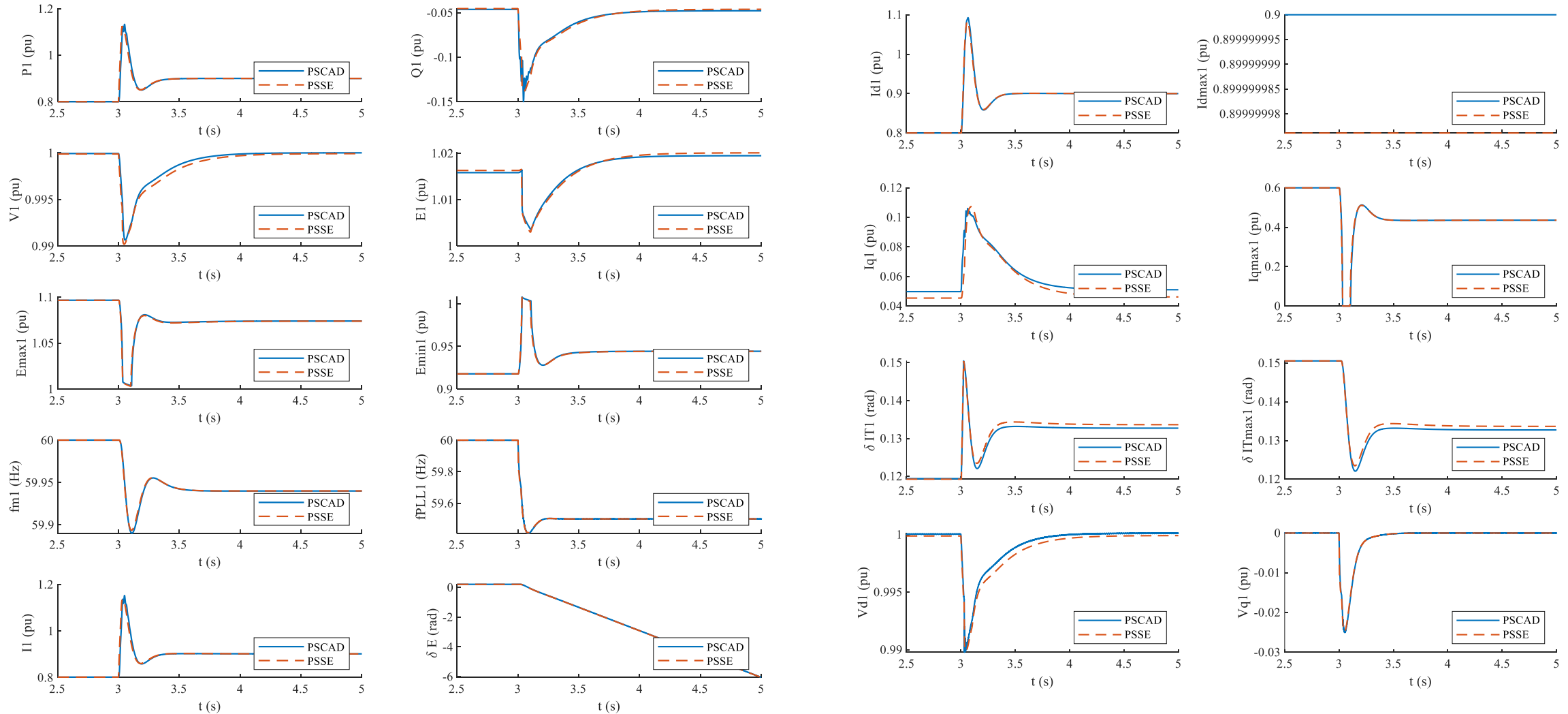
- The VSM GFM responds to the frequency change by increasing P autonomously
- The frequency drop results in the overload of VSM GFM
- I_d was limited by limiting the angle δ_{IT}



Active current I_d limiting

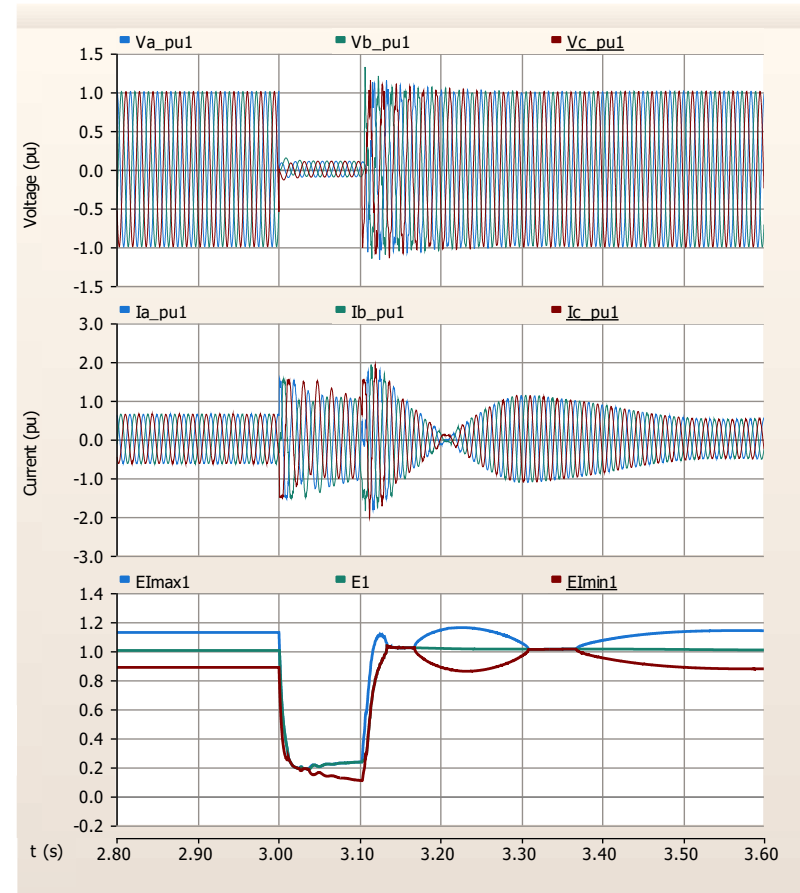
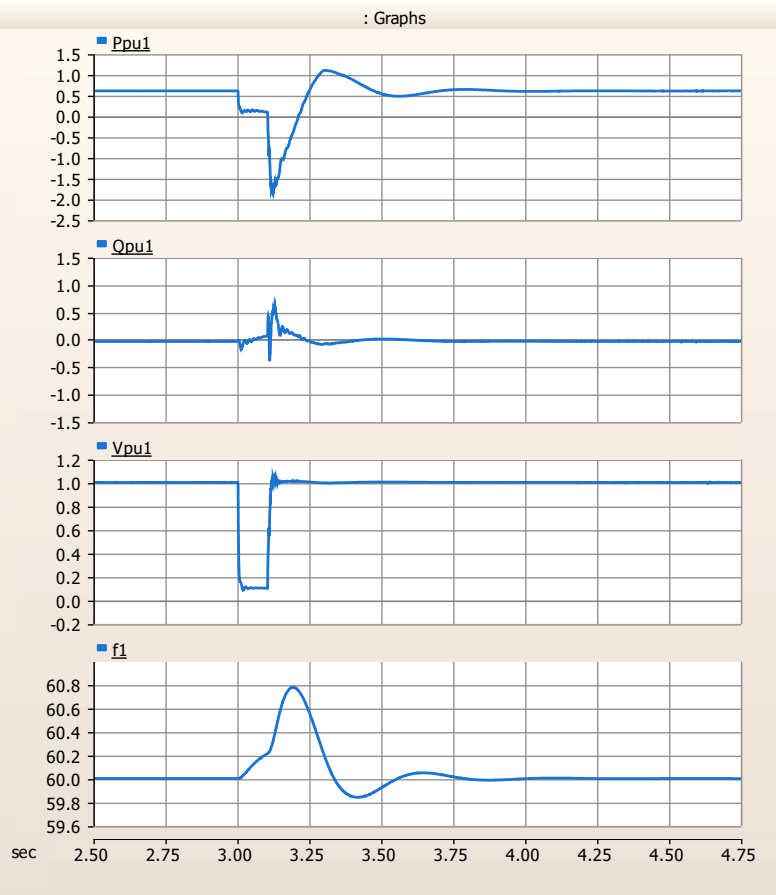
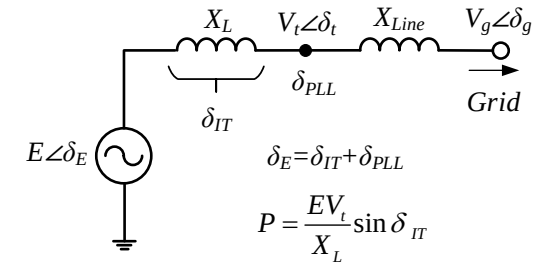
Frequency Drop (PSS/E and PSCAD comparison)

- 18 variables were compared and achieved a good level of alignment



0.1 s fault

- The transient current limiting clips the current at I_{maxF} 1.5 pu
- The reactive current limiting (E_{max} and E_{min}) later limits the current at 1 pu
- The active current limiting also plays a role in limiting the steady-state current



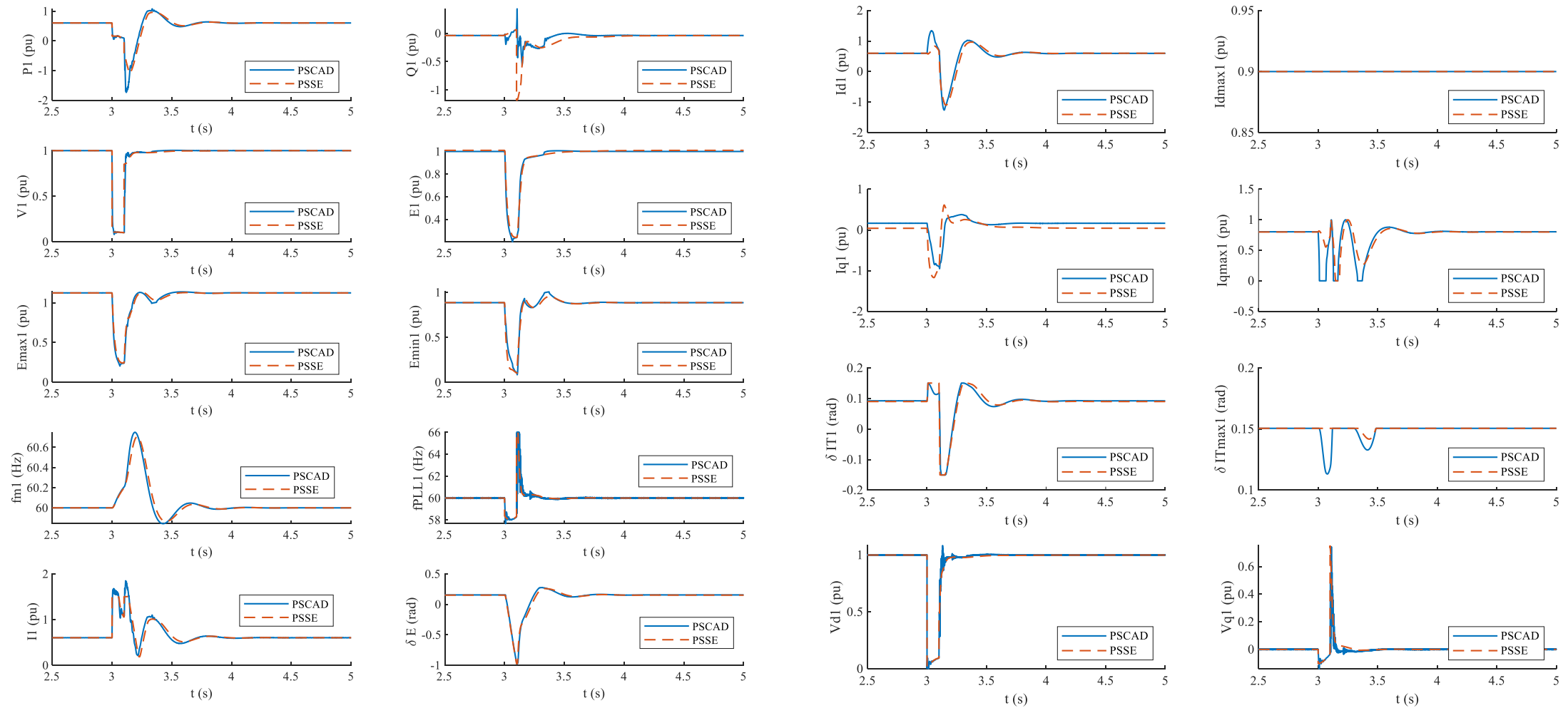
$$E_{min} = \sqrt{(V_{inv} - I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

$$E_{max} = \sqrt{(V_{inv} + I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

Reactive current limiting

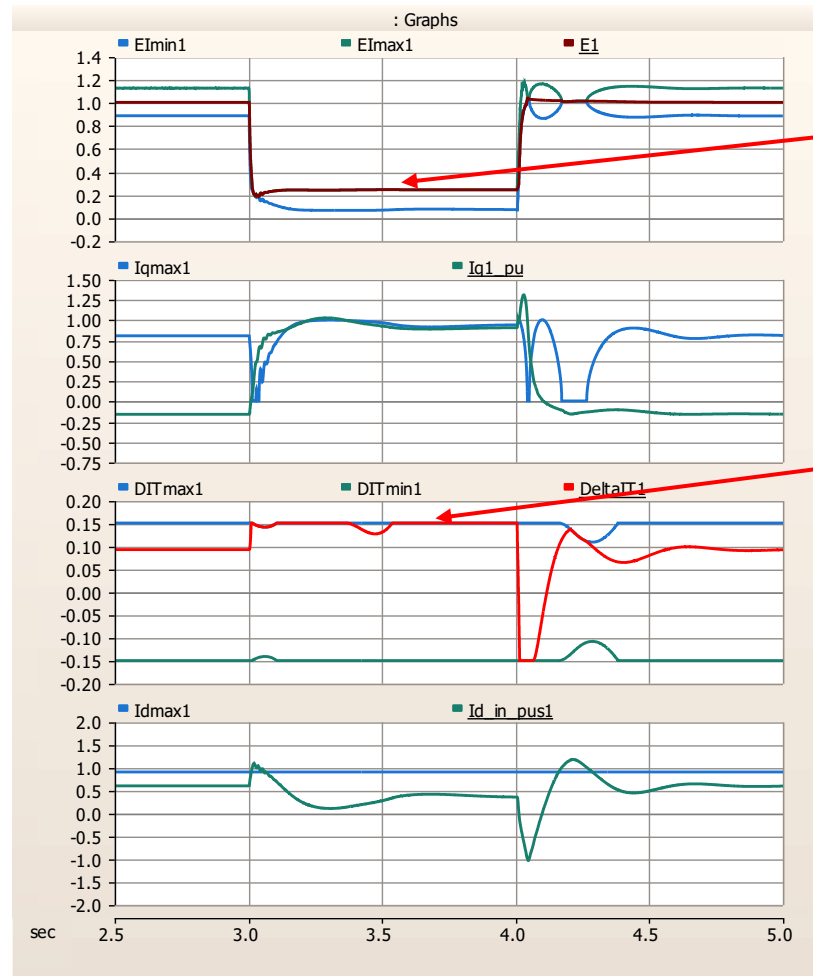
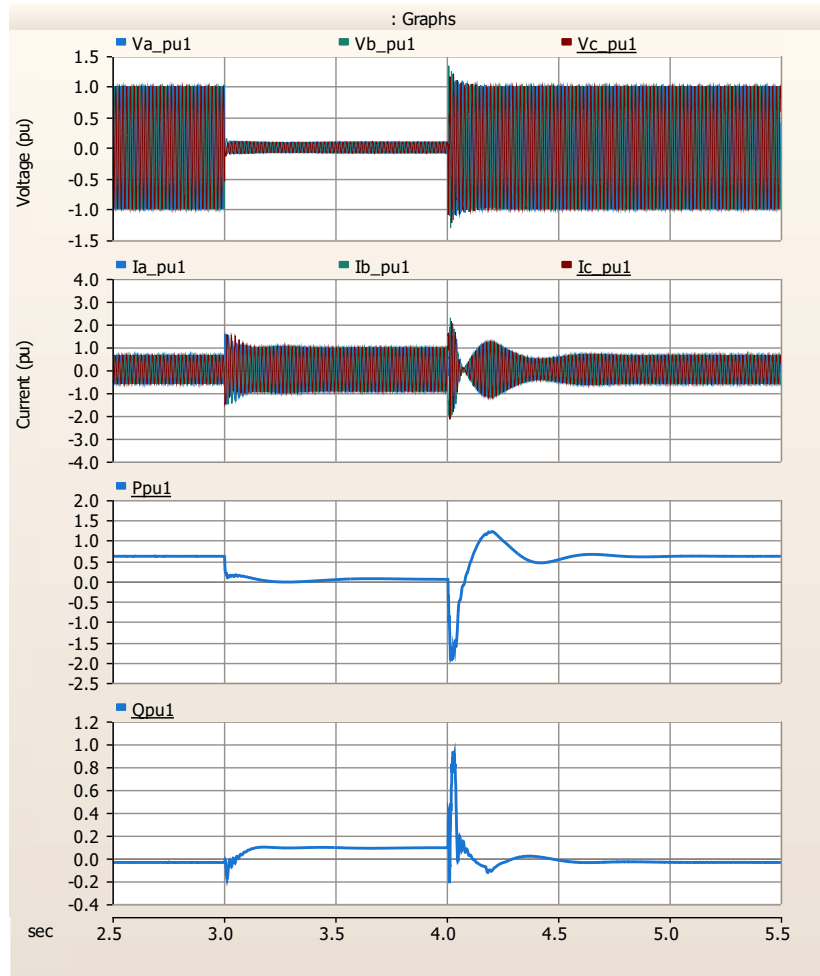
0.1 s fault (PSS/E and PSCAD comparison)

- 18 variables were compared and achieved a good level of alignment



1 s fault

- The transient current limiting clips the current at $I_{maxF} = 1.5$ pu
- The steady-state current limiting later limits the current at $I_{maxSS} = 1$ pu by reducing E and clamping phase angle
- Initial overcurrent post fault because of reduced E

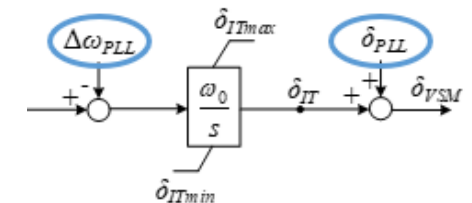


I_q is limited by E_{max}/E_{min} controller

$$E_{min} = \sqrt{(V_{inv} - I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

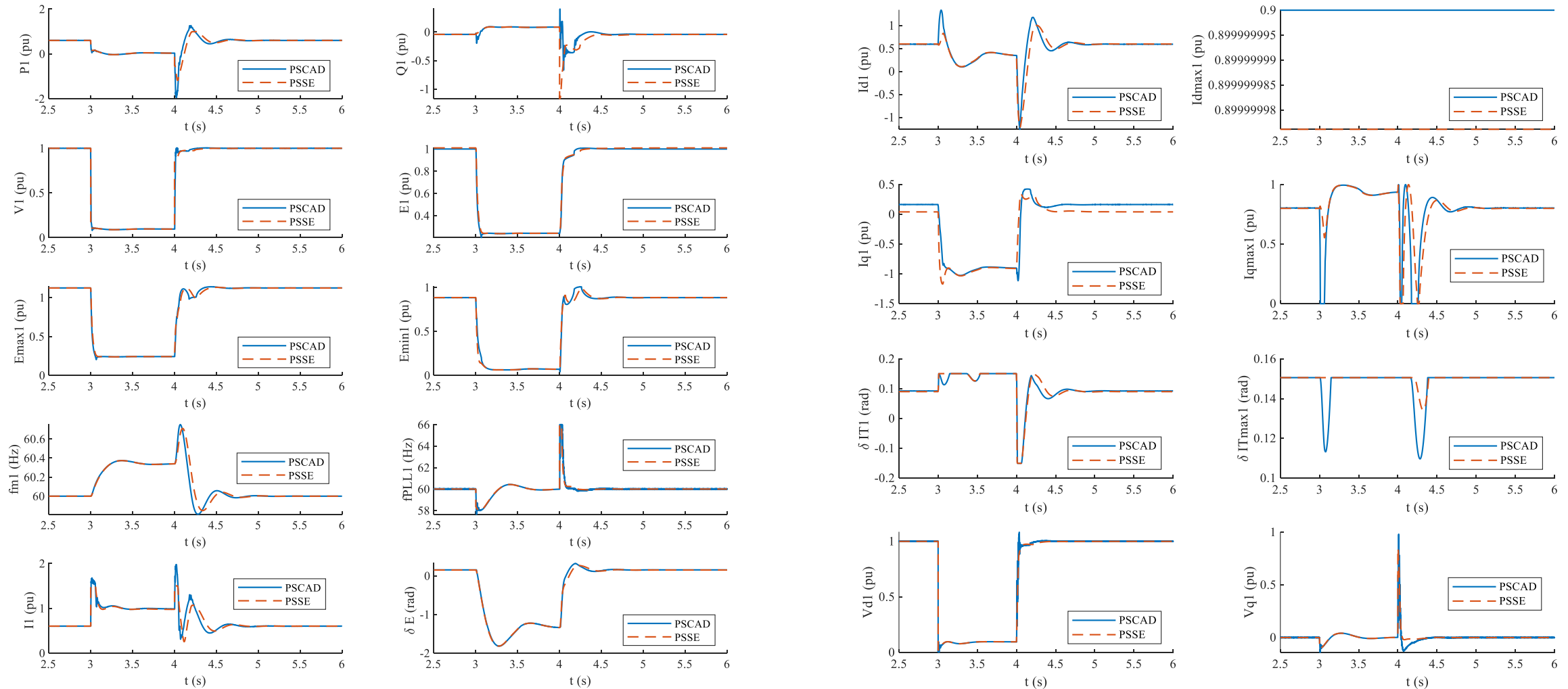
$$E_{max} = \sqrt{(V_{inv} + I_{qmax} X_L)^2 + (I_{dinv} X_L)^2}$$

I_d is limited by the saturation limits δ_{ITmax} and δ_{ITmin}



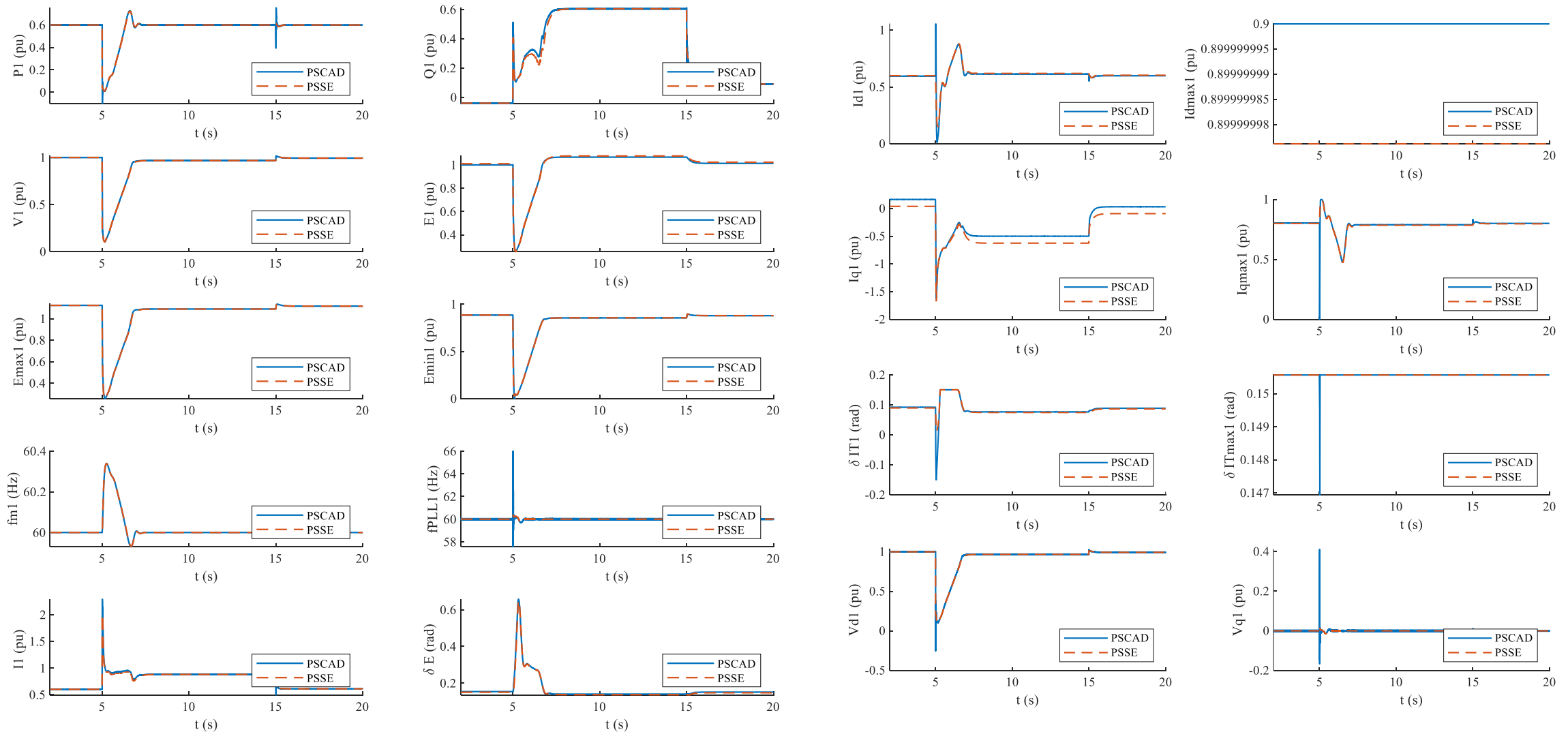
1 s fault (PSS/E and PSCAD comparison)

- 18 variables were compared and achieved a good level of alignment



ERCOT LVRT Test [1]

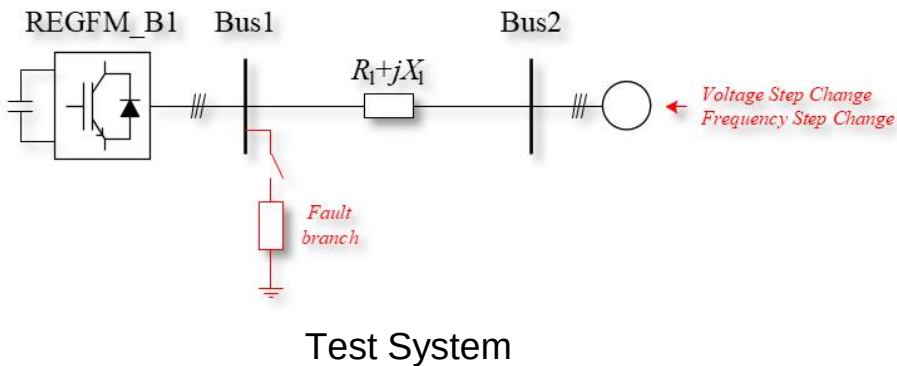
- 18 variables were compared and achieved a good level of alignment



[1] Yunzhi Cheng, <https://sites.google.com/view/dmview/home>

REGFM_B1 Model Benchmarking Results

- The model was benchmarked in a single-GFM-infinite-bus system, and different tests including voltage step change, frequency step change, low-impedance and high-impedance faults were compared
- Different flag combinations were also considered

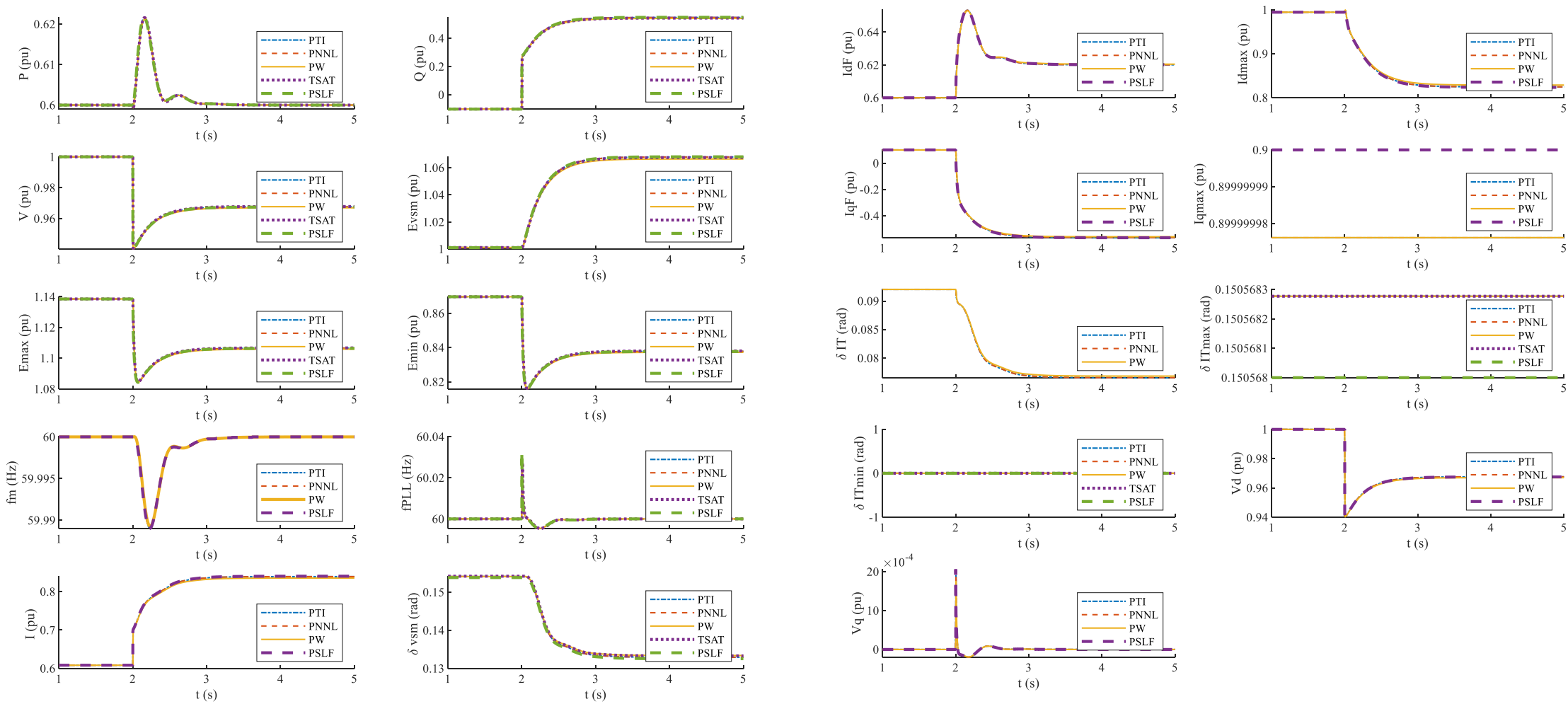


Parameters

symbol	Value	symbol	Value
R_s	0.02 pu	D_2	100 pu
X_s	0.15pu	ω_d	50 pu
m_q	0.01 pu	I_{maxss}	1 pu
k_{pv}	0.00 pu	k_{pf}	0.9
k_{iv}	5.86 pu	k_i	2 pu
m_p	0.01 pu	I_{maxf}	2 pu
$d\omega_{max}$	0.1 pu	T_{pf}	0.02 s
$d\omega_{min}$	-0.1 pu	T_{qf}	0.02 s
k_{pPLL}	0.265 pu	T_{vf}	0.02 s
k_{iPLL}	2.650 pu	T_{if}	0.02 s
$d\omega_{PLLmax}$	0.1 pu	k_e	1
$d\omega_{PLLmin}$	-0.1 pu	V_{base}	230 kV
T_p	0.02 pu	M_{base}	100 MVA
H	3 s	R_1	0.02 pu
D_1	0.001 pu	X_1	0.10 pu

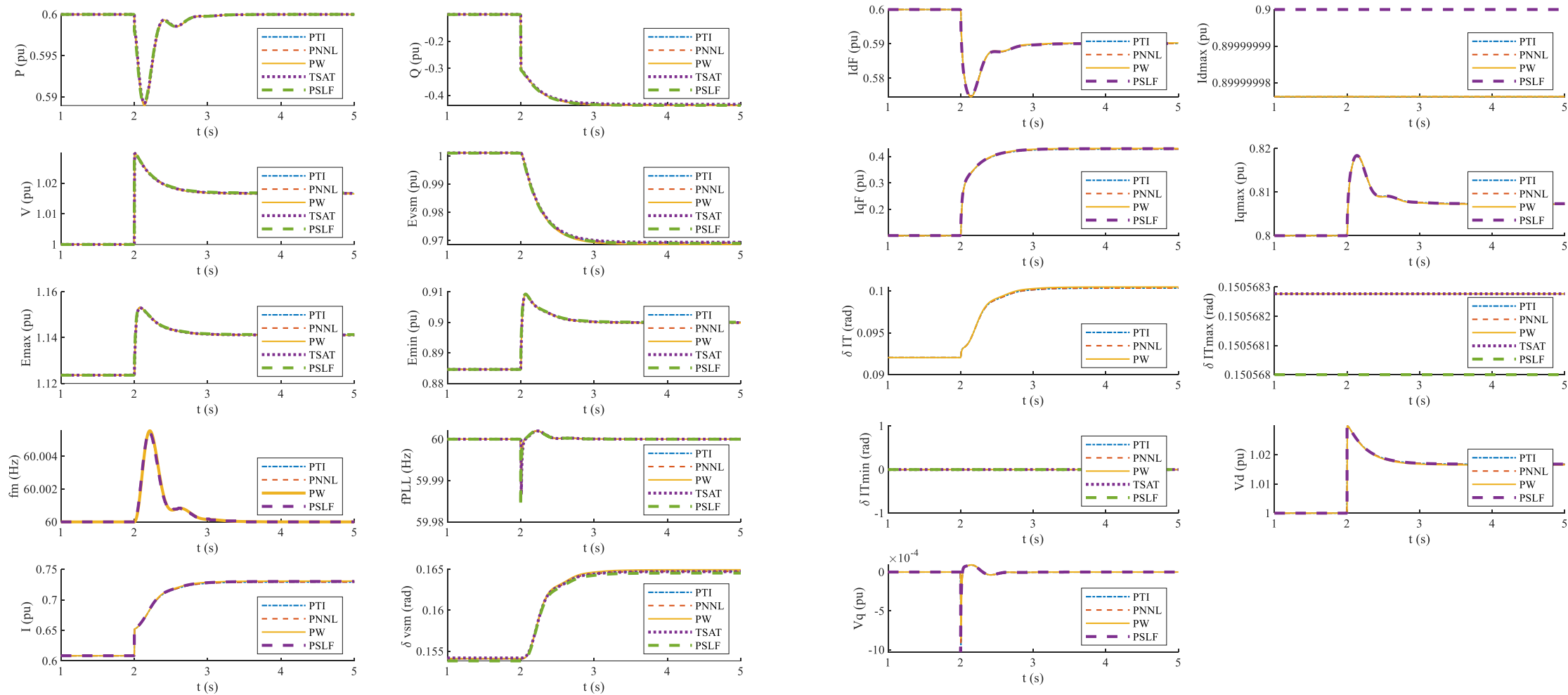
Voltage step down by 0.1 pu

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	0	0	0	0



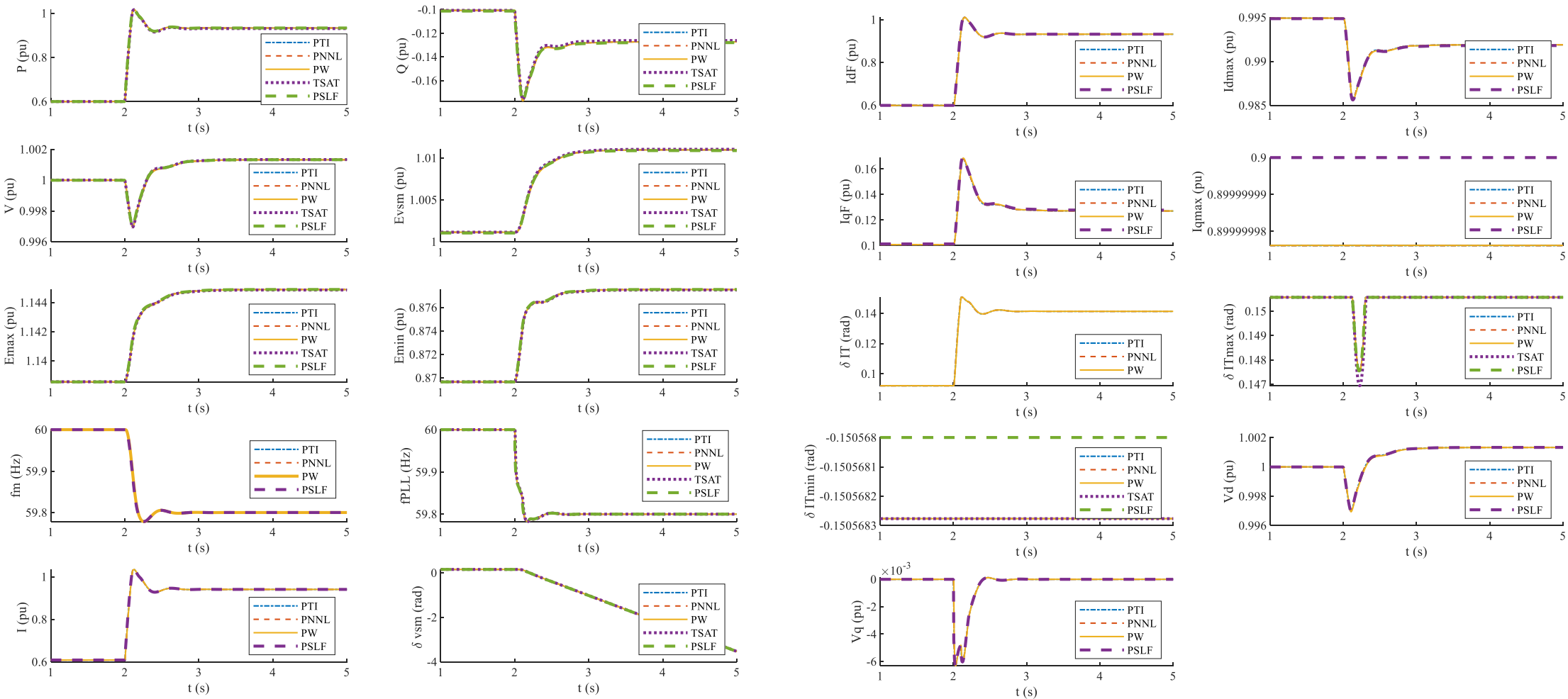
Voltage step up by 0.05pu

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	0	1	0	0



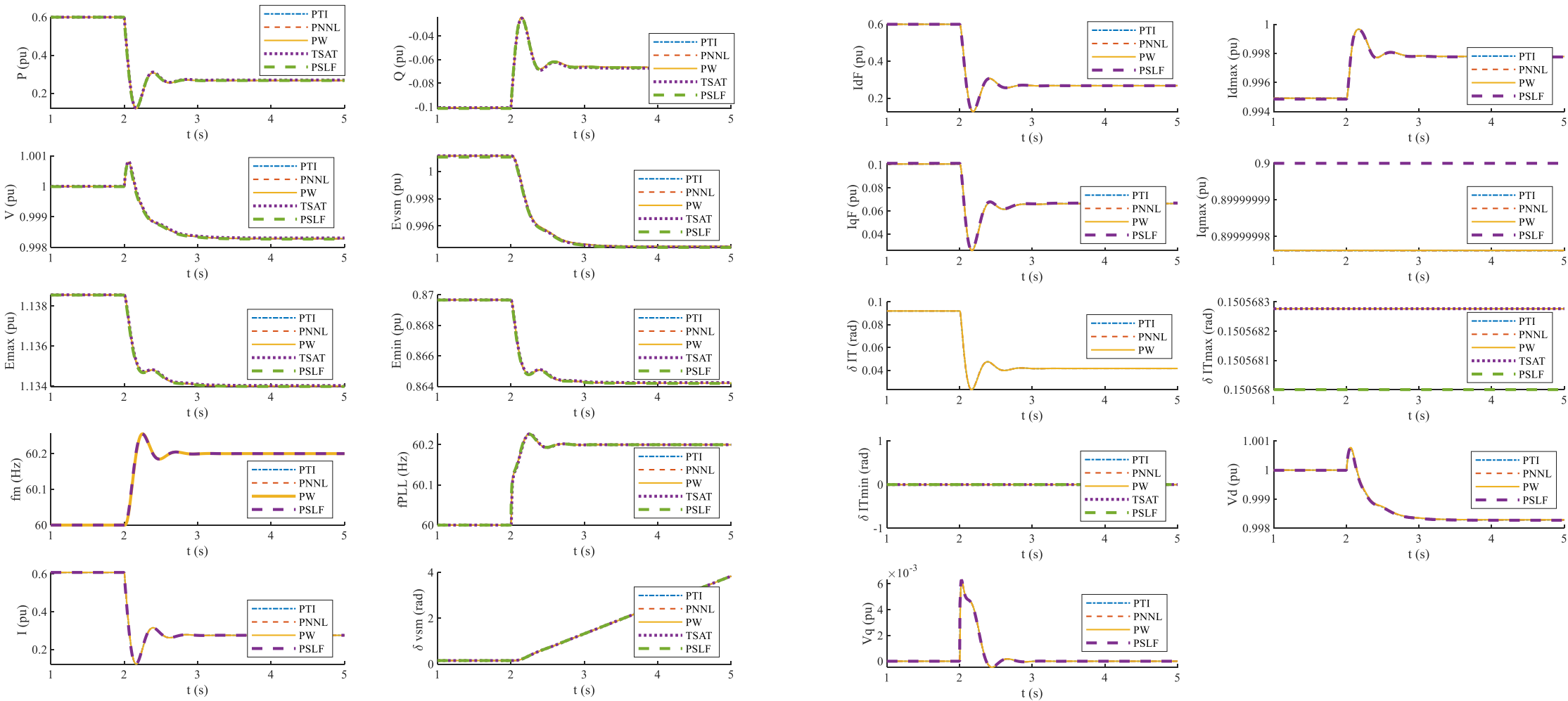
Frequency step down to 59.8 Hz

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	1	0	0	0



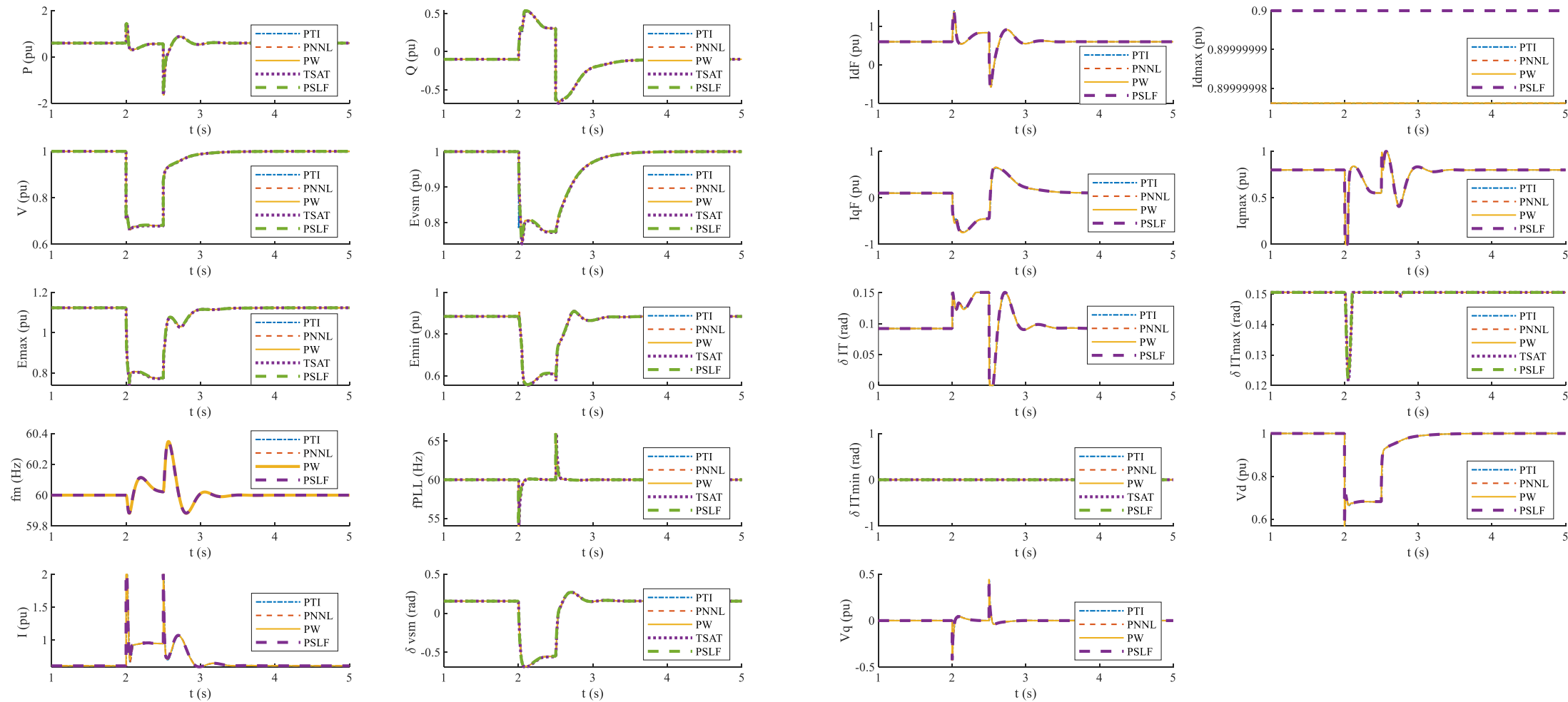
Frequency step up to 60.2 Hz

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	0	0	0	0



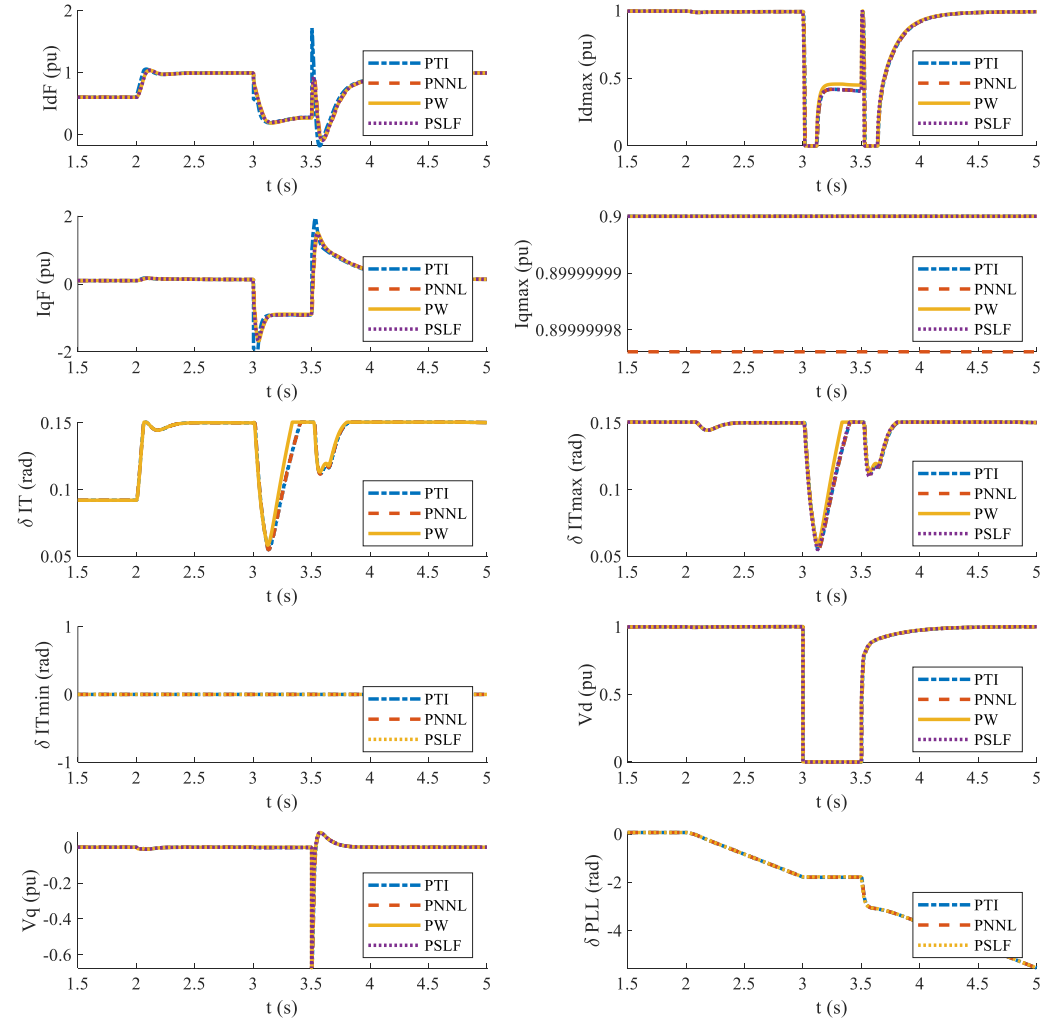
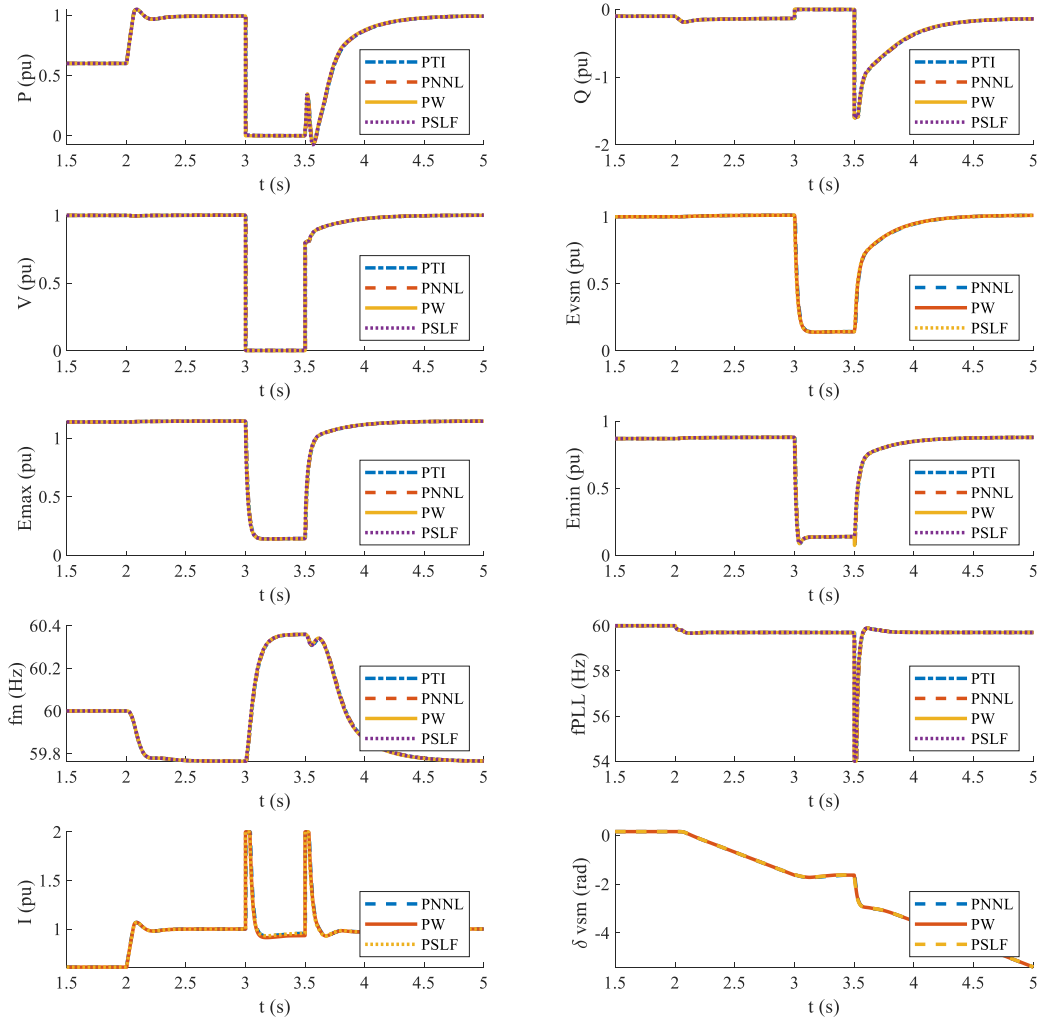
Fault 50 Ohm for 0.5s

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	0	1	0	0



0.1 Ohm Fault at 59.7 Hz

F Flag (Fixed)	Vdrp Flag (Fixed)	ES Flag	PQ Flag	QV Flag	ω Flag
1	0	0	1	0	0



Conclusions

- The REGFM_B1 specification was approved by WECC MVS on **Jan. 25th, 2024**
- The REGFM_B1 model has been implemented in PSS/E, PSLF, PowerWorld, and TSAT, and the model benchmarking is complete in **May 2024**
- The REGFM_B1 model can reflect the main characteristics of a virtual synchronous machine grid-forming inverter

I'd like to make a motion to approve the REGFM_B1 model



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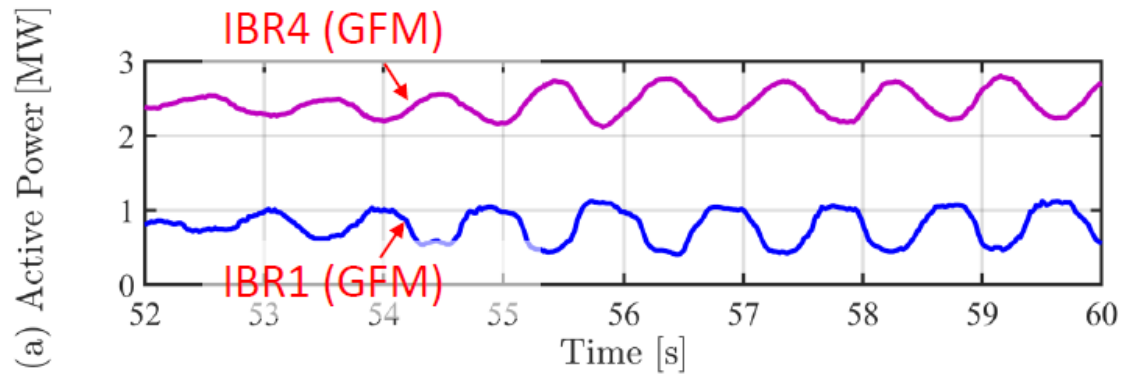
This material is based upon work supported by the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) under Solar Energy Technologies Office (SETO) Award Number 38637. The views expressed herein do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

Wei Du

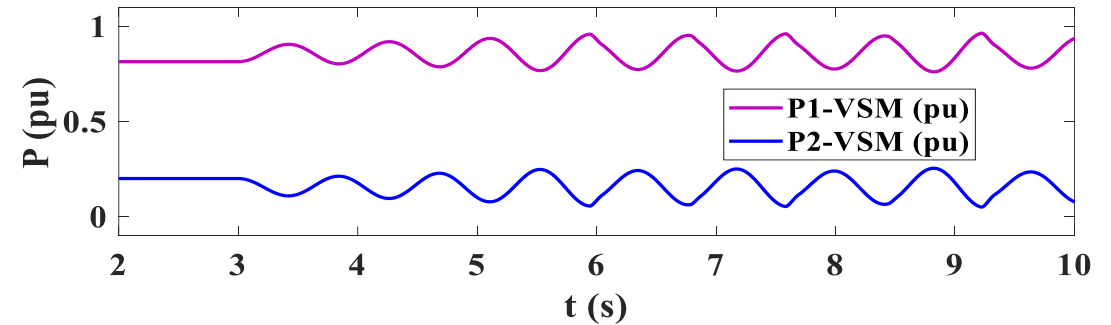
Wei.du@pnnl.gov

THANK YOU

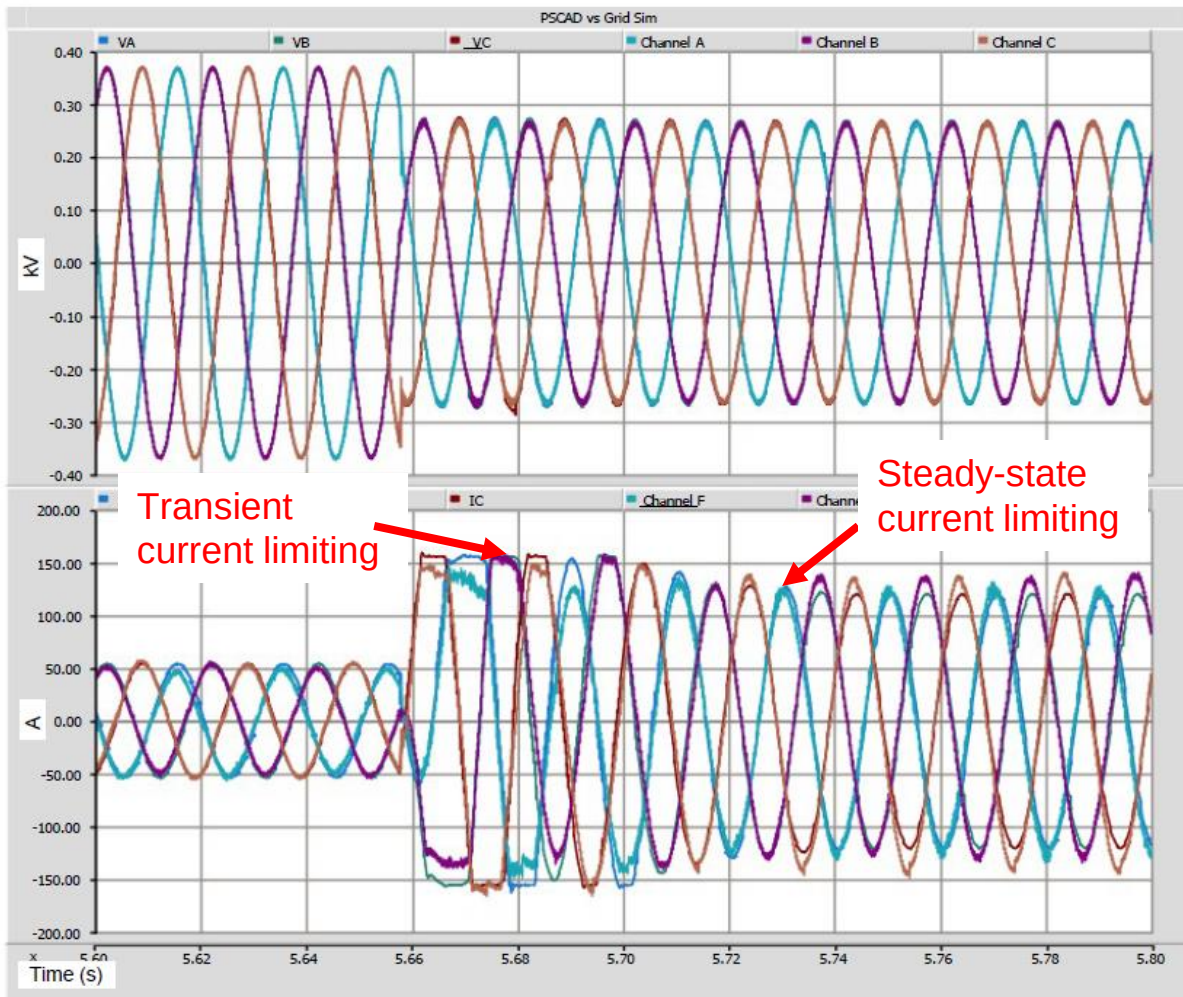
The REGFM_B1 model can reproduce the 1 Hz oscillation in Hawaiian Island



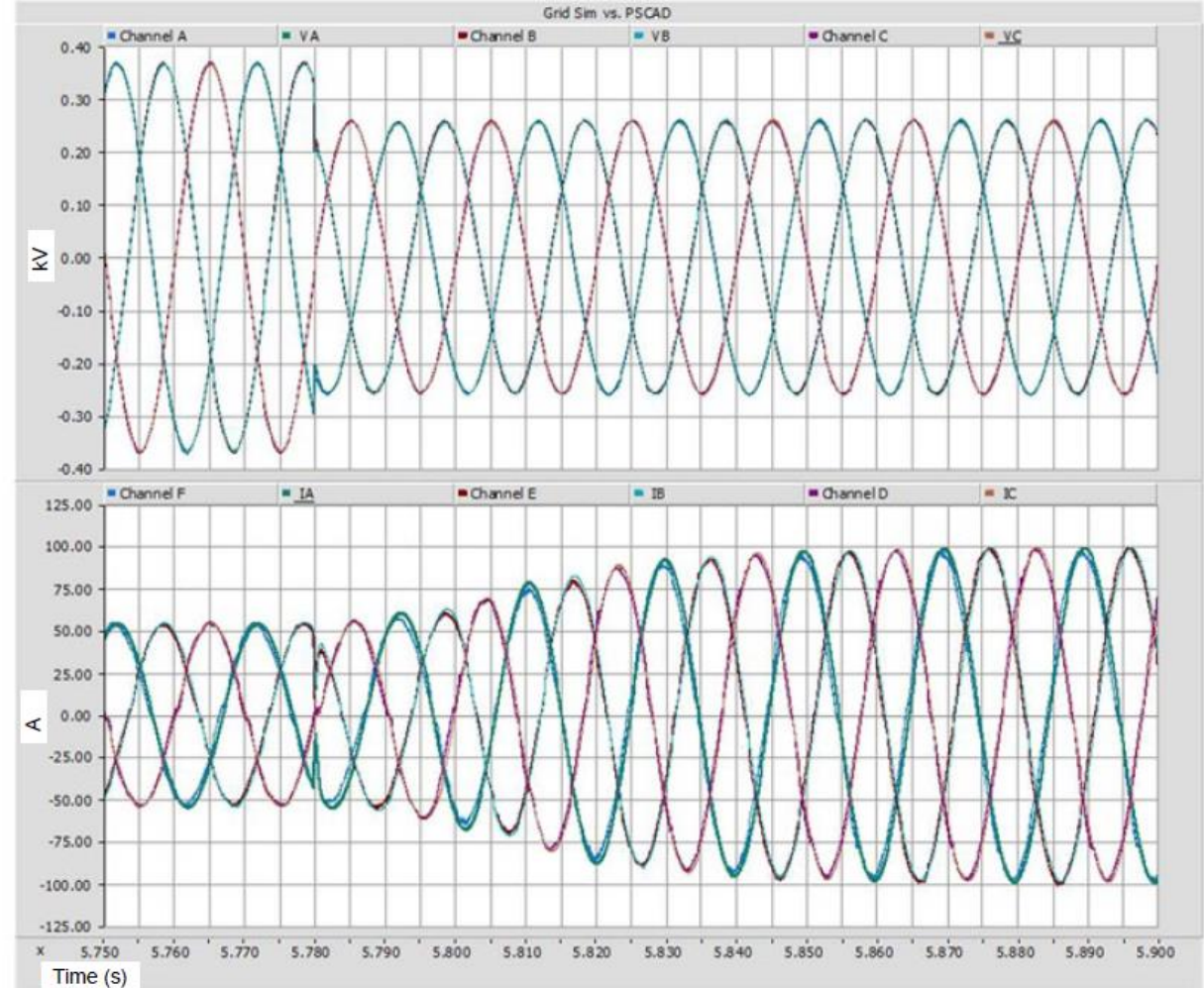
Field results. Event: on Apr. 30th, 2023, two GFMs oscillate against each other, and the oscillation frequency is 1 Hz.
(Source: Jin Tan, NREL)



The REGFM_B1 model can reproduce the 1 Hz oscillation.



OEM simulation results of a GFM response to a 0.3 pu voltage drop [1]



OEM simulation results of a GFL response to a 0.3 pu voltage drop [1]

[1] <https://arena.gov.au/knowledge-bank/hornsedale-power-reserve-virtual-machine-mode-testing-summary-report/>